

Paper ID: 16301

Roll No.....

END-SEMESTER EXAMINATION, 2023-24**B. Tech. (SEMESTER : 03)****CSC 201 : DIGITAL FORENSICS****Time: 3 Hrs.****Max. Marks: 100**

- Note:** 1. All questions are compulsory.
2. Assume missing data suitably, if any.

- CO1: Demonstrate the principles of Digital Forensics and how resultant evidence can be applied within legal cases.
CO2: Illustrate their competence in recovering files, network forensics, password cracking
CO3: Evaluate the effectiveness of available digital forensics tools and use them in a way that optimizes the efficiency and quality of digital forensics investigations.
CO4: Apply a solid foundational grounding in computer networks, operating systems, file systems, hardware, and mobile devices to digital investigations and to the protection of computer network resources from unauthorized activity
CO5: Access and critically evaluate relevant technical and legal information and emerging industry trends
CO6: Adapt effectively the results of a computer, network, and/or data forensic analysis verbally, in writing, and in presentations to both technical and lay audiences.

COs Marks BTL**SECTION-A****All Questions are Compulsory:****(10×4=40 Marks)**

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|--|-----|---|----|
| 1. What are the 5 rules of Digital Evidence? | CO1 | 4 | K1 |
| 2. Explain the process of preserving and collecting digital evidence. How does it differ from physical evidence in a crime scene? | CO1 | 4 | K2 |
| 3. What types of digital evidence can be collected through live forensics techniques? | CO2 | 4 | K1 |
| 4. Explain the importance of minimizing data contamination during live evidence collection. | CO2 | 4 | K2 |
| 5. Explain working of file recovery using Recuva. | CO3 | 4 | K2 |
| 6. Choose case study of your choice to understand the need of external Hardware based Write Blocker. | CO3 | 4 | K3 |
| 7. Explain the working of Ethical Hacker. | CO4 | 4 | K2 |
| 8. Identify the meaning of Black Box Testing. | CO4 | 4 | K3 |
| 9. Classify types of hackers. | CO5 | 4 | K2 |
| 10. Choose any four and write Short notes on these protocols:
a. Ipv4 and Ipv6 b. IMAP c. VoIP d. P2P d. TCP/IP | CO5 | 4 | K3 |

SECTION-B**All Questions are Compulsory:****(3×6=18 Marks)**

11. (a) Examine the types of digital data and storage media that can be subject to a data forensics investigation. CO3 6 K4

--- OR ---

- (b) Inspect what are the steps for searching data on Windows Based system.
12. (a) Analyse the types of information and data that can be extracted from routers during a forensic investigation. CO4 6 K4

--- OR ---

- (b) Discover the Forensics Process and explain each phases / steps.
13. (a) Examine the common techniques and tools used in email forensics to recover deleted emails, trace email senders, and verify the authenticity of email content. CO5 6 K4

--- OR ---

- (b) Compare and contrast between email spamming and email bombing.

SECTION-C

All Questions are Compulsory:

(3×10=30 Marks)

14. (a) Analyse the best practices for ensuring the integrity and chain of custody when using hardware tools for data acquisition. CO3 10 K4/K5

--- OR ---

- (b) Compare and contrast between Network Based Firewall and Web Application Firewall.

15. (a) Compare and contrast between TCP and UDP poisoning. CO4 10 K4/K5

--- OR ---

- (b) Analyse the common types of wireless attacks. How do they pose security risks to wireless networks?

16. (a) What are the various types of web attacks? Compare any three of them. CO5 10 K4/K5

--- OR ---

- (b) Explain various types of Cybercrimes. Tabulate and classify them with examples in detail.

SECTION-D

Attempt the following Question:

(1×12=12 Marks)

17. (a) Define Steganography. How it is different from Cryptography? List the various types of steganography with the approaches used to carry them out. Describe one of the steganography techniques with an example. CO6 12 K5/K6

--- OR ---

- (b) Elaborate on the working of the write blocker with a suitable diagram.

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